

QQ: A Language Metadata Toolkit for Multilingual NLP

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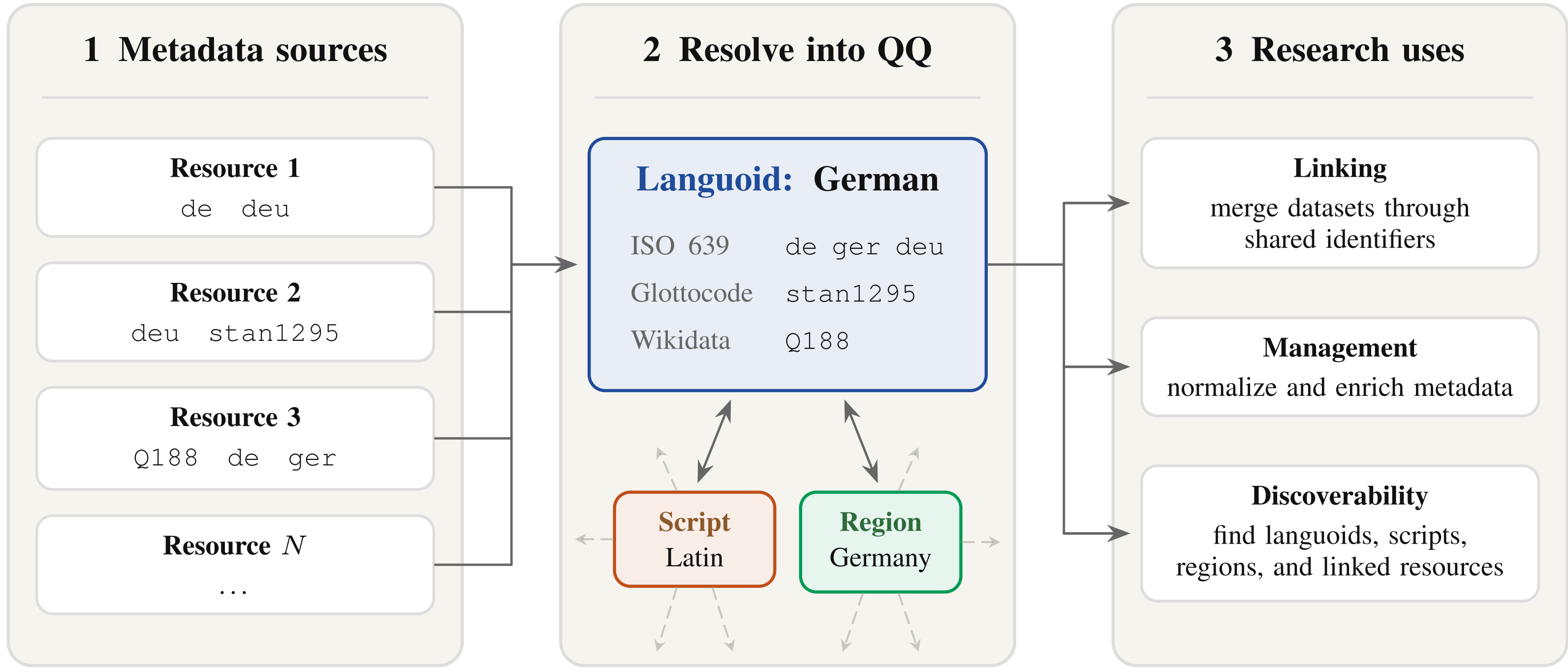
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German

de deu ger
deu_Latn stan1295 Q188

many identifiers × many languages



1. Language Metadata

- Multilingual NLP now spans **thousands of languages** across **many datasets**
- The same language can have many identifiers
- One-off scripts or mappings are annoying and error-prone

Goal: Make language metadata management easier.

2. What Is QQ?

QQ merges open metadata sources into a **graph** database.

- **Nodes:** languoids (languages, dialects, and families), scripts, and regions
- **Metadata:** names, identifiers, families, scripts, speakers, status, and resource links
- Qwanqwa (ዳንዳ) means “language” in Amharic; QQ is nice and short to type

FAIR: Searchable, versioning, open, available through PyPI, GitHub, and Zenodo.

One Database, Three Interfaces

Python	Normalize, query, join, traverse
CLI	Shell workflows
Explorer	Search, inspect the graph, discover resources

Python

```
from qq import Database, IdType
```

```
db = Database.load()
dutch = db.guess("dutc1256")
```

```
dutch.name
# "Dutch"
dutch.endonym
# "Nederlands"
dutch.glottocode
# "dutc1256"
dutch.wikidata_id
# "Q7411"
dutch.speaker_count
# 24085200
dutch.endangerment_status
# "Not endangered"
dutch.level
# "language"
dutch.name_in("fr")
# "Néerlandais"
```

```
db.convert("nl-Latn-BE", IdType.ISO_639_3)
# "nld"
assert dutch == db.guess("nl") == db.guess("nld_Latn")
# True
```

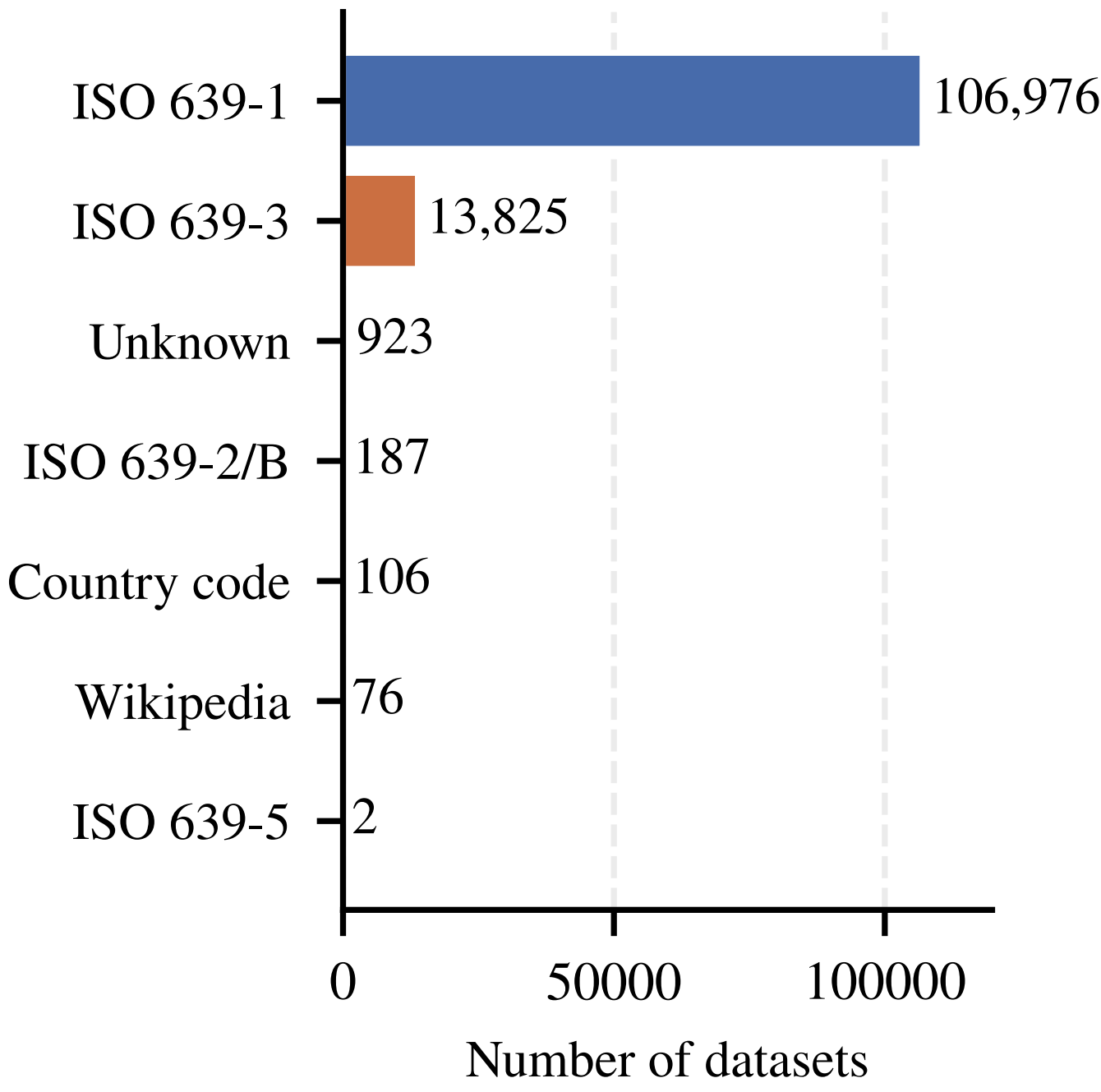
Web Interface

The screenshot shows the 'About this explorer' page with a search bar. Below, the 'Bavarian' language entry is displayed, including its name, endonym (Boarisch), description, and identifiers (BCP-47: bar, ISO 639-1: deu, ISO 639-2B: deu, ISO 639-3: bar). A graph view shows the relationship between Bavarian and other languages like Central Bavarian, East Central Bavarian, and South Bavarian. The 'Resources' section lists datasets like Hugging Face and Universal Dependencies.

3. Workflow Evaluation

Hugging Face Audit

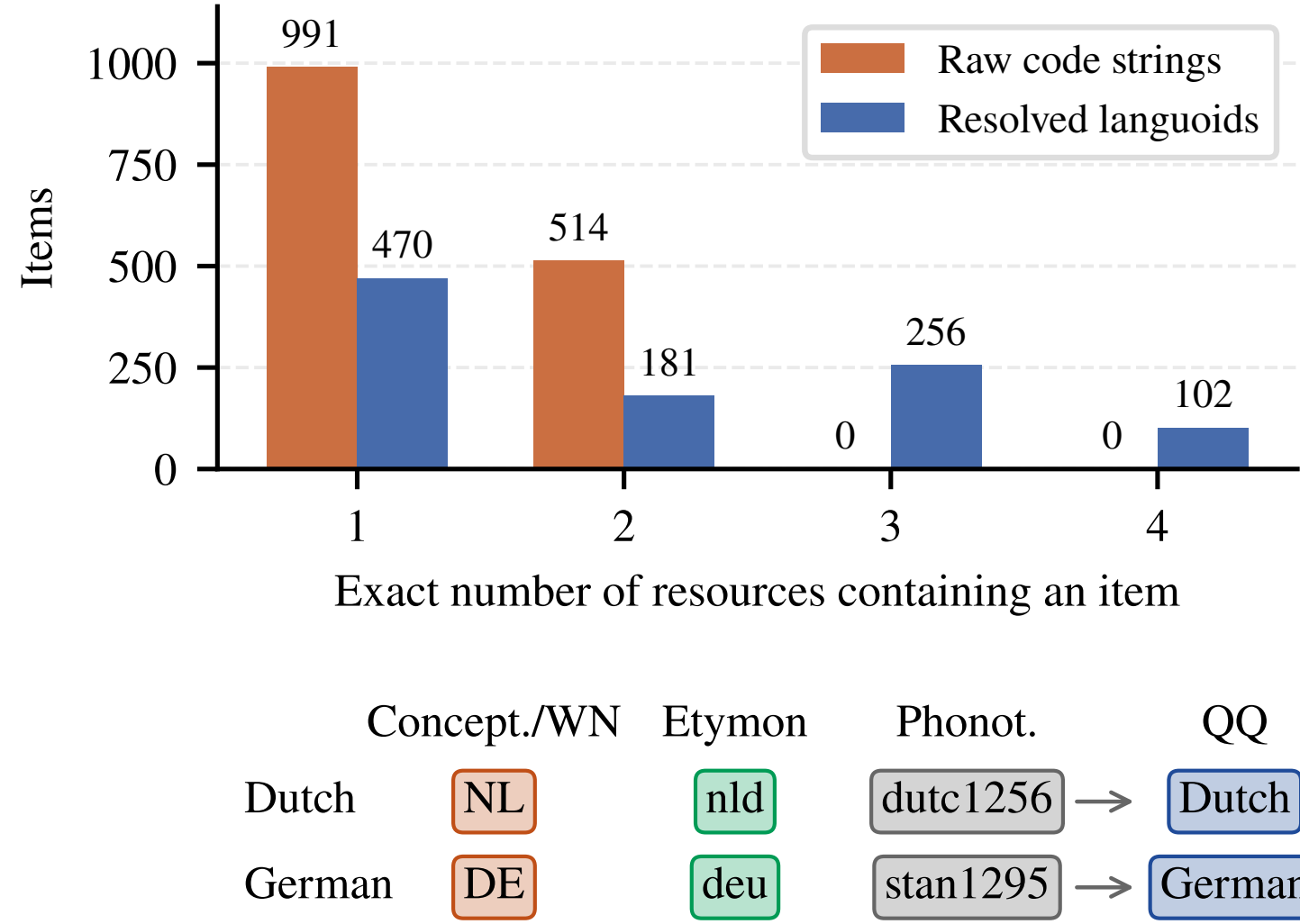
Scan all HF datasets metadata for language tags.



Audit result: QQ resolved **98.9%** of unique tags: **8,144** of **8,237**, representing **7,943** languoids.

Linking Resources

Normalize identifiers across different resources.



Linking result: **102 languages** are shared by all four resources after normalization.

Traverse the Graph

Family relations

```
dutch.parent
# "Global Dutch"
dutch.parent.parent
# "Modern Dutch"
dutch.family_tree
# [..., "Germanic", ..., "Indo-European"]
dutch.siblings
# ["Afrikaansic", "Javindo", "Petjo", ...]
dutch.descendants
# ["North Hollandish", ...]
```

Multi-hop discovery

```
db.get_region("ET").name
# "Ethiopia"
db.get_script("Ethi").name
# "Ethiopic"
```

```
amharic = db.get("am")
amharic.languoids_in_same_region
# ["Xamtanga", "Dorze", ...]
amharic.languoids_with_same_script
# ["Wolane", "Saho", "Bilin", ...]
```

4. Takeaways

One database, three interfaces: Python, CLI, and browser
Normalize: Reuse results for auditing, linking, discovery, and reporting
Reproducible: Versioned database and open formats (FAIR)

Install

pip install qwanqwa

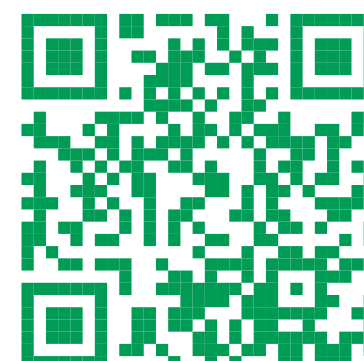
Try QQ



Explorer
wesselpoelman.nl/qq



GitHub
github.com/WPoelman/qwanqwa



Paper
arxiv.org/abs/2603.00620

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